Sentinel-2 Multiclass Land-Cover & Cloud Image Classification

Pixel vs Patch-Based Deep Learning Pipeline for Satellite Image Analysis

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Problem Setting

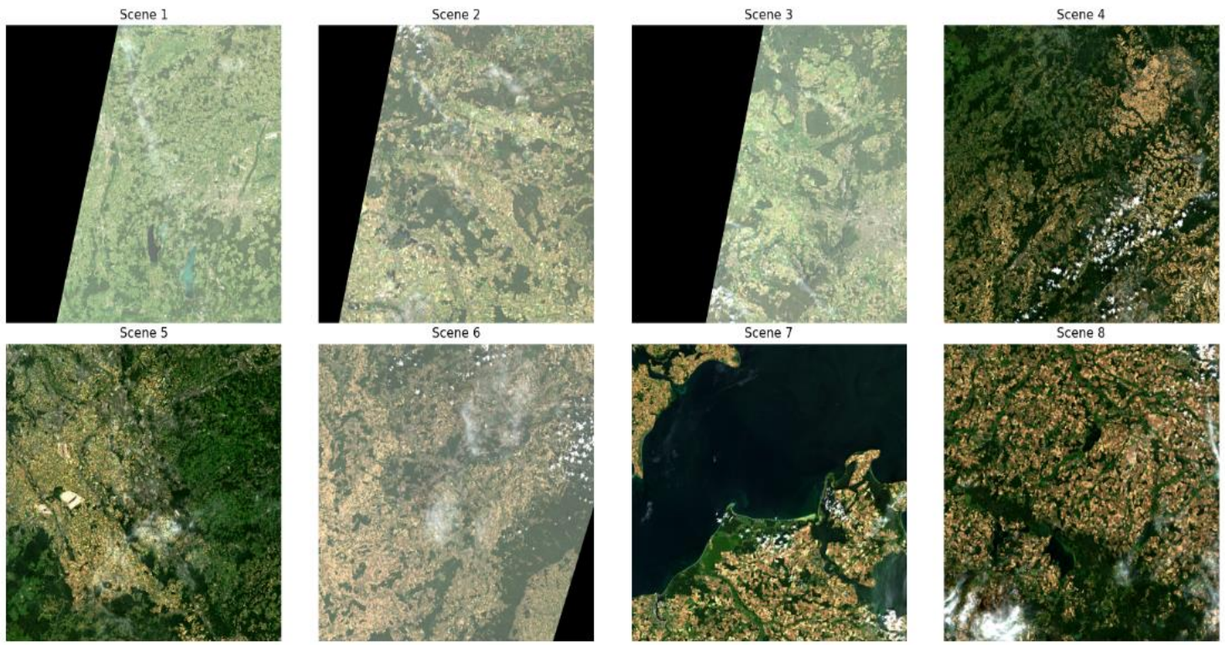
Task: Supervised multiclass classification of Sentinel-2 images

Dataset: 8 Sentinel-2 scenes, 10 spectral bands (20 m), ~213 M labeled pixels

Classes: Vegetation (56.2%), Bare Soil (18.2%), Water (8.9%), Cloud (16.6%)

Challenges:

- Class skewness: Vegetation dominates, Water/Cloud underrepresented
- Spectral similarity: Vegetation vs Clouds overlap
- Cloud contamination: Optical data affected by atmosphere



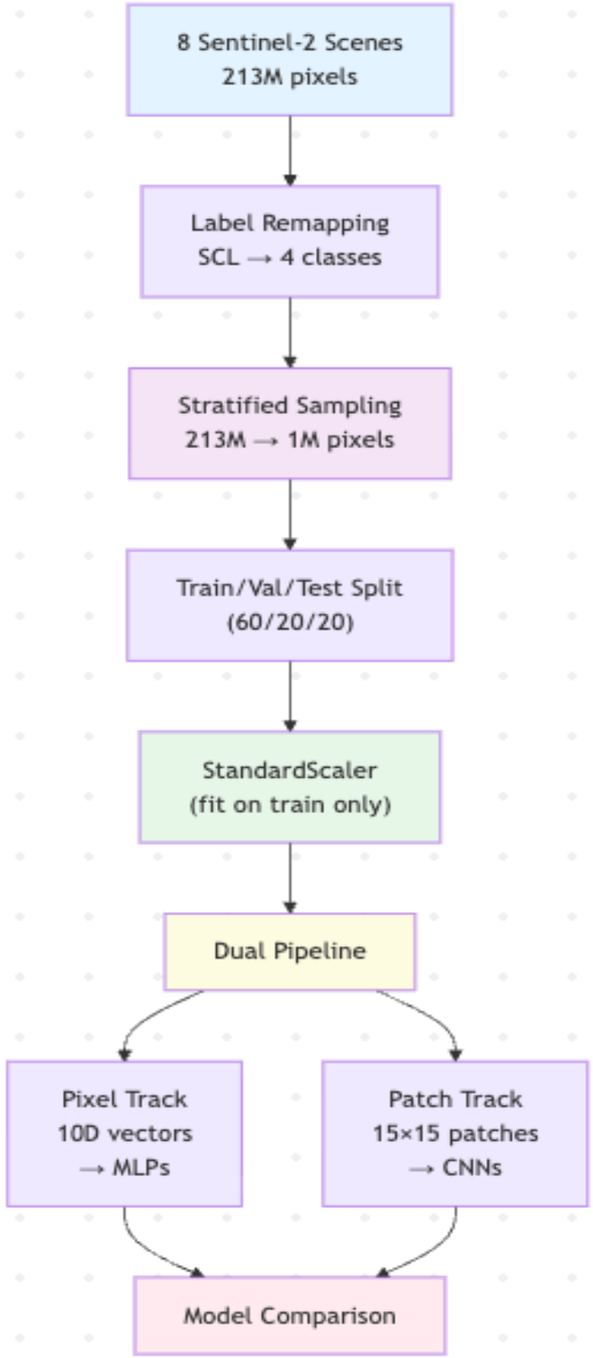
Preprocessing Pipeline

Two-track approach for architectural comparison:

- Pixel-based (MLPs): 213M→1M sampled pixels, 10-band spectral vectors, stratified sampling
- Patch-based (CNNs): Same 1M sampled Pixels → 15×15 spatial patches, boundary-safe extraction (reflective padding), memory-mapped storage

Common setup: Train/Val/Test = 60/20/20, StandardScaler fitted on training only

```
Class %: Original → Sampled, with counts
Vegetation: 56.21% → 56.21%  | counts: orig 119,762,936 vs samp 562,134
 Bare Soil: 18.22% → 18.22%  | counts: orig 38,823,931 vs samp 182,228
     Water:  8.93% →  8.93%  | counts: orig 19,029,195 vs samp 89,318
     Cloud: 16.63% → 16.63%  | counts: orig 35,434,581 vs samp 166,320
```



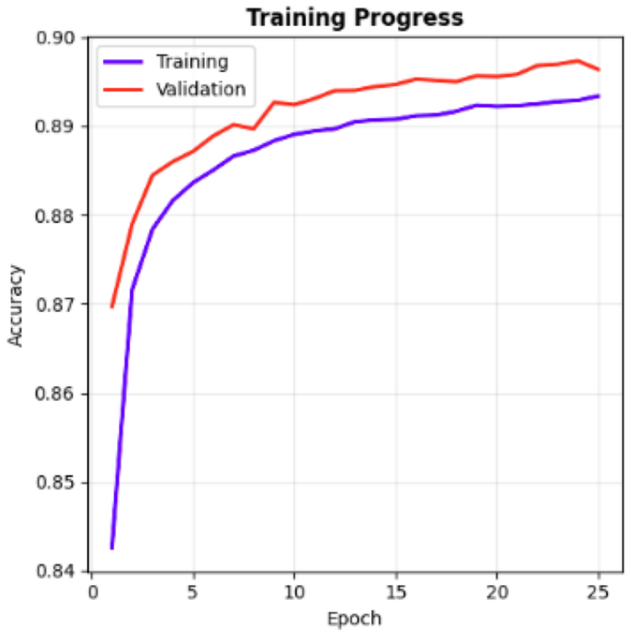
Model 1 – Baseline MLP

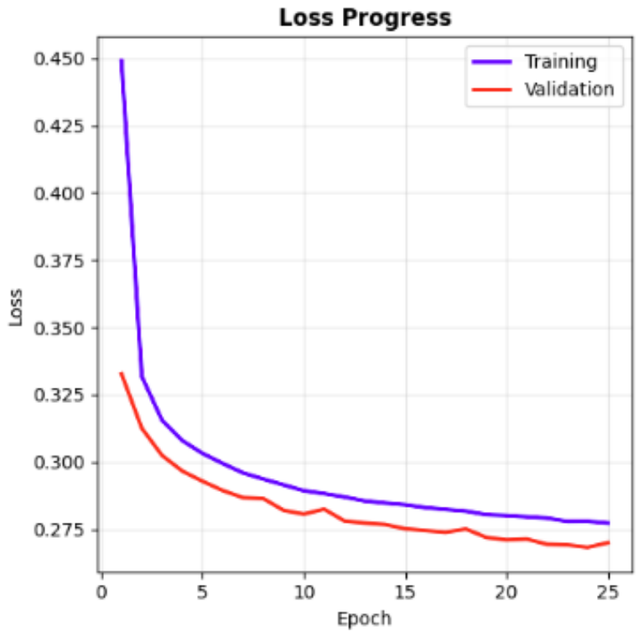
```
BASELINEMLP ARCHITECTURE
--------------------------------------------------
BaselineMLP(
  (network): Sequential(
    (0): Linear(in_features=10, out_features=64, bias=True)
    (1): ReLU()
    (2): Dropout(p=0.2, inplace=False)
    (3): Linear(in_features=64, out_features=32, bias=True)
    (4): ReLU()
    (5): Linear(in_features=32, out_features=4, bias=True)
  )
)
```

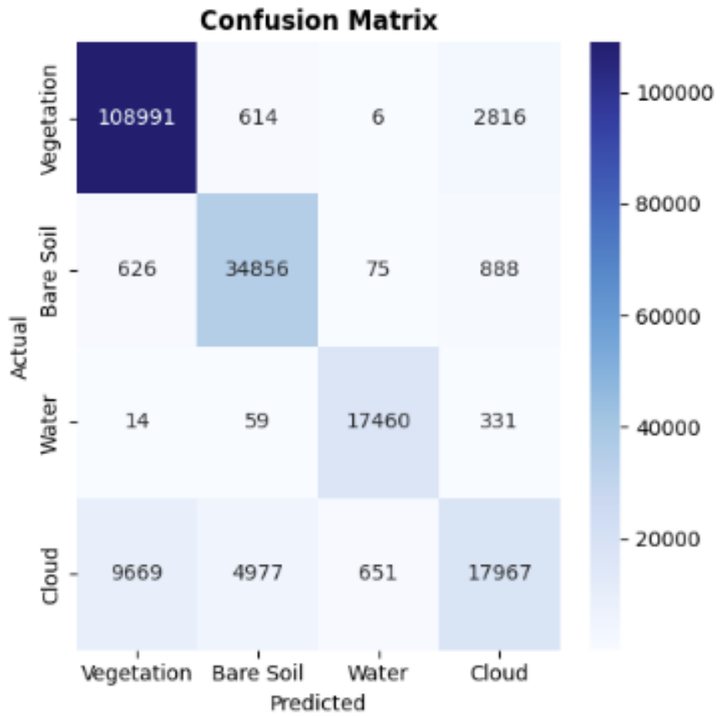
Parameters: 2,916
Best Validation Accuracy: 0.8973
Total Training Epochs: 25

DETAILED CLASSIFICATION REPORT:

	precision	recall	f1-score	support
Vegetation	0.9136	0.9694	0.9407	112427
Bare Soil	0.8605	0.9564	0.9059	36445
Water	0.9598	0.9774	0.9685	17864
Cloud	0.8166	0.5401	0.6502	33264
accuracy			0.8964	200000
macro avg	0.8876	0.8608	0.8663	200000
weighted avg	0.8919	0.8964	0.8885	200000





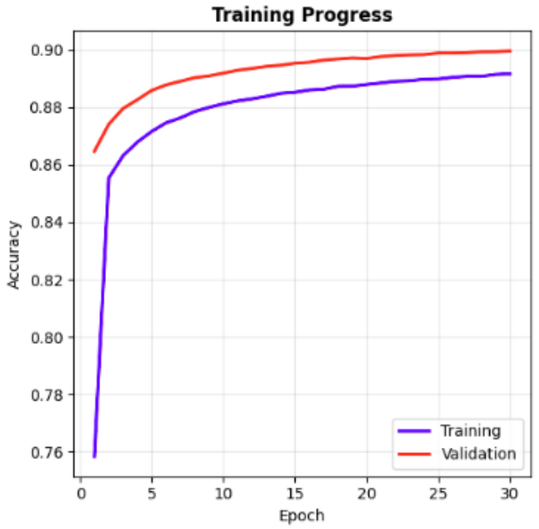


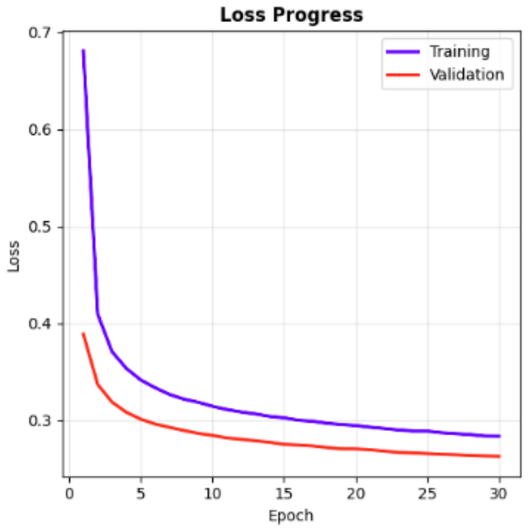
Model 2 – Advanced MLP

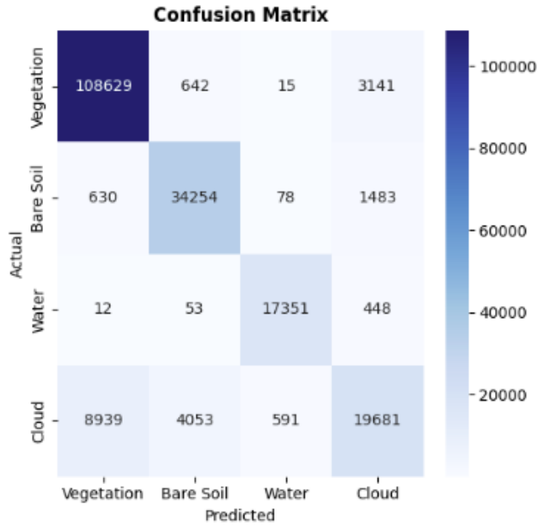
```
DYNAMICMLP ARCHITECTURE
--------------------------------------------------
DynamicMLP(
  (layers): ModuleList(
    (0): Linear(in_features=10, out_features=256, bias=True)
    (1): Linear(in_features=256, out_features=128, bias=True)
    (2): Linear(in_features=128, out_features=64, bias=True)
  )
  (batch_norms): ModuleList(
    (0): BatchNorm1d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (1): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  )
  (dropouts): ModuleList(
    (0): Dropout(p=0.3, inplace=False)
    (1): Dropout(p=0.27, inplace=False)
    (2): Dropout(p=0.243, inplace=False)
  )
  (output_layer): Linear(in_features=64, out_features=4, bias=True)
  (activation_fn): ReLU(inplace=True)
)
```

```
DETAILED CLASSIFICATION REPORT:
              precision    recall  f1-score   support

  Vegetation     0.9662    0.9189    0.9420    118210
   Bare Soil     0.9399    0.8783    0.9080     39002
       Water     0.9713    0.9621    0.9667     18035
       Cloud     0.5917    0.7951    0.6785     24753

    accuracy                         0.8996    200000
   macro avg     0.8673    0.8886    0.8738    200000
weighted avg     0.9152    0.8996    0.9050    200000

KEY INSIGHTS:
--------------------
Cloud recall: 0.592 (improved from baseline)
Architecture improvement: 45,124 parameters vs 2,276 in baseline
Advanced MLP analysis complete!
Improvement over Baseline MLP: +0.0032
```

```
Model Architecture: 10 → 256 → 128 → 64 → 4
Parameters: 45,124
Best Validation Accuracy: 0.8996
Total Training Epochs: 30
```

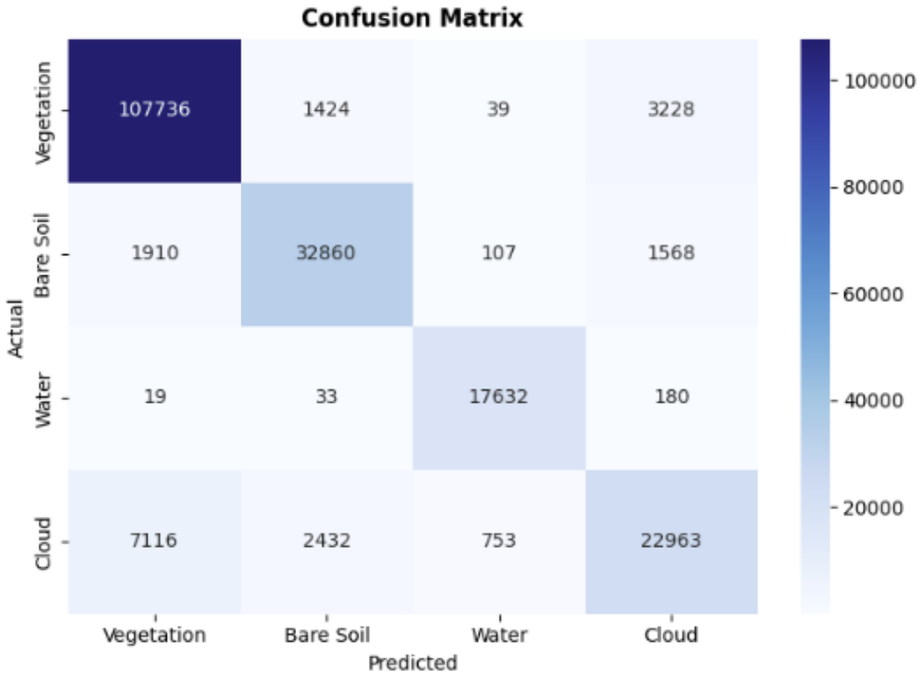




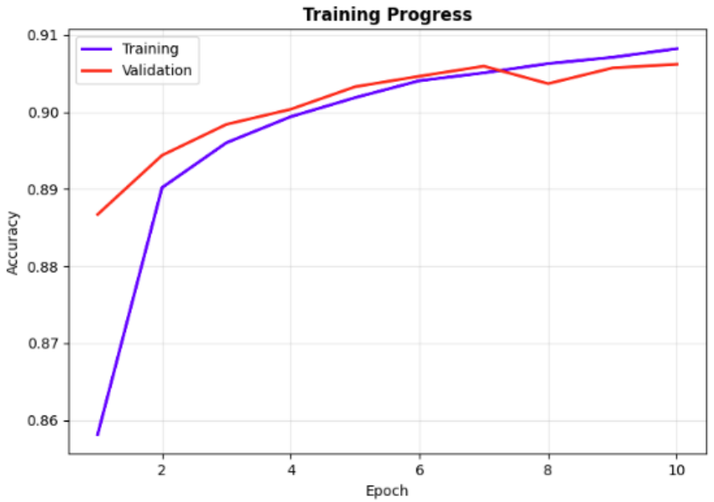


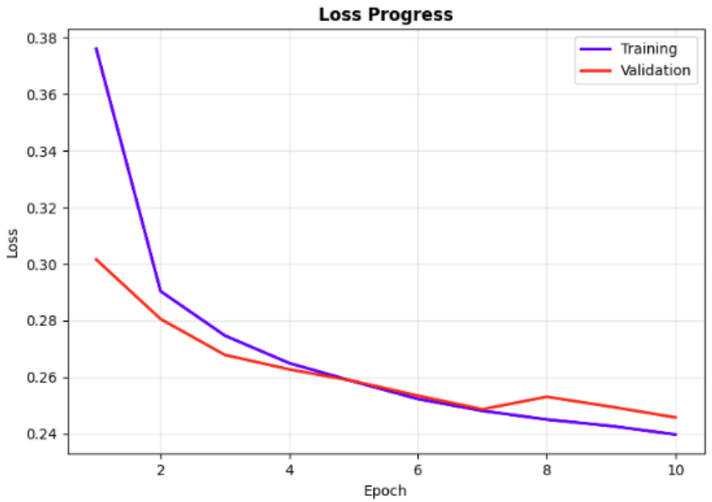
Model 3 – Basic CNN

```
BASIC CNN ARCHITECTURE
--------------------------------------------------
BasicCNN(
  (conv): Conv2d(10, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (pool): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
  (fc1): Linear(in_features=3136, out_features=16, bias=True)
  (fc2): Linear(in_features=16, out_features=4, bias=True)
)

Model Architecture: Conv → MaxPool → Flatten → FC16 → FC4
Parameters: 56,084
Best Validation Accuracy: 0.9060
Total Training Epochs: 10

VALIDATION SET PERFORMANCE:
Accuracy: 0.9060
F1 (macro): 0.8893
```




```
DETAILED CLASSIFICATION REPORT:
              precision    recall  f1-score   support

  Vegetation     0.9225    0.9583    0.9401    112427
   Bare Soil     0.8942    0.9016    0.8979     36445
       Water     0.9515    0.9870    0.9689     17864
       Cloud     0.8219    0.6903    0.7504     33264

    accuracy                         0.9060    200000
   macro avg     0.8975    0.8843    0.8893    200000
weighted avg     0.9032    0.9060    0.9034    200000
```



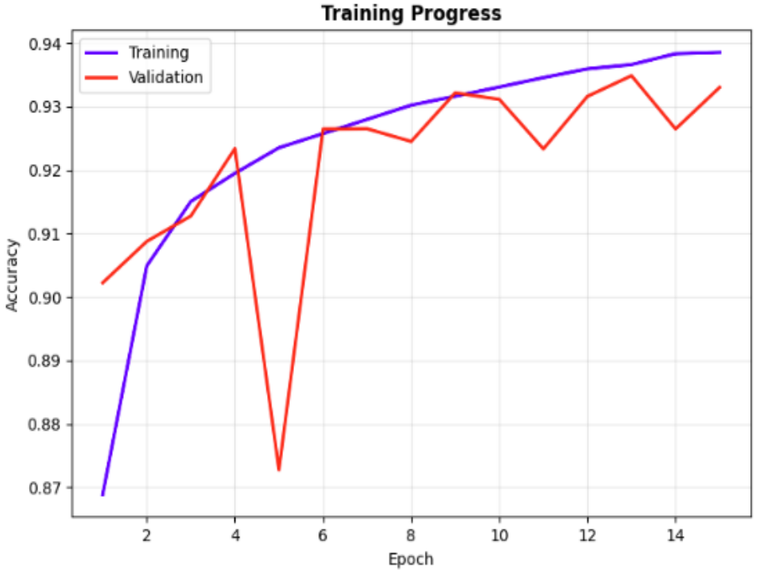


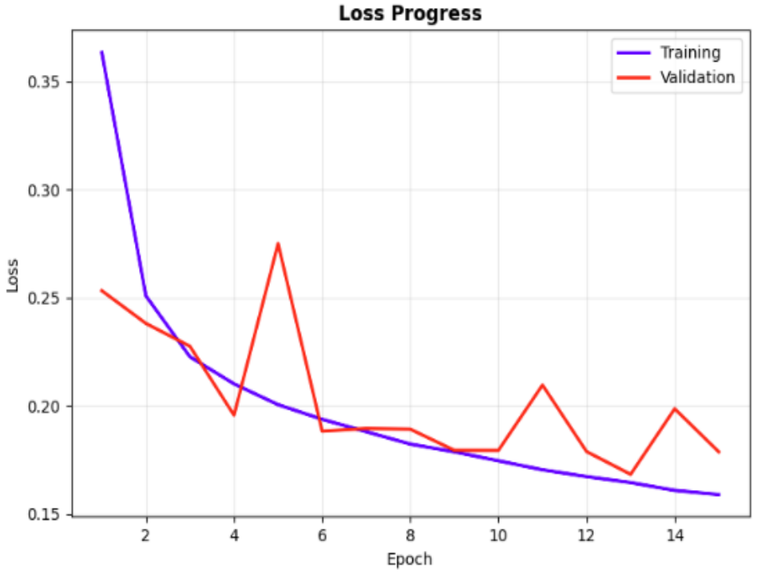
Model 4 – Advaned CNN

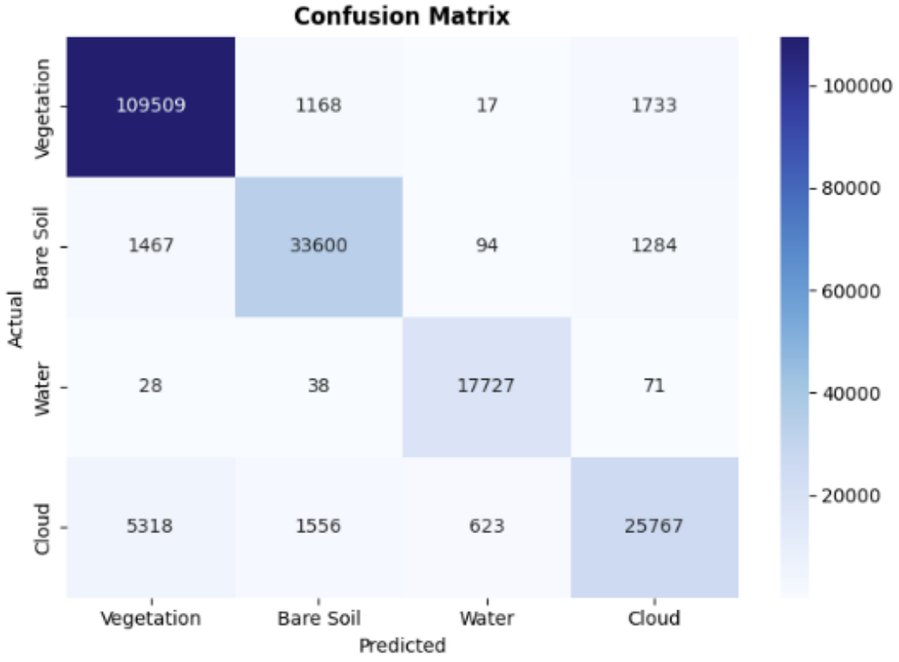
```
ADVANCED CNN - PERFORMANCE ANALYSIS
============================================================
Model Architecture: 3 Conv Blocks (32→64→128) + Global Pooling
Parameters: 150,276
Best Validation Accuracy: 0.9349
Total Training Epochs: 15

VALIDATION SET PERFORMANCE:
Accuracy:      0.9330
```

```
DETAILED CLASSIFICATION REPORT:
              precision    recall  f1-score   support

  Vegetation     0.9414    0.9740    0.9575    112427
   Bare Soil     0.9240    0.9219    0.9230     36445
       Water     0.9602    0.9923    0.9760     17864
       Cloud     0.8930    0.7746    0.8296     33264

    accuracy                         0.9330    200000
   macro avg     0.9297    0.9157    0.9215    200000
weighted avg     0.9319    0.9330    0.9316    200000
```

- Substantial improvements (+2.89% over Basic CNN)



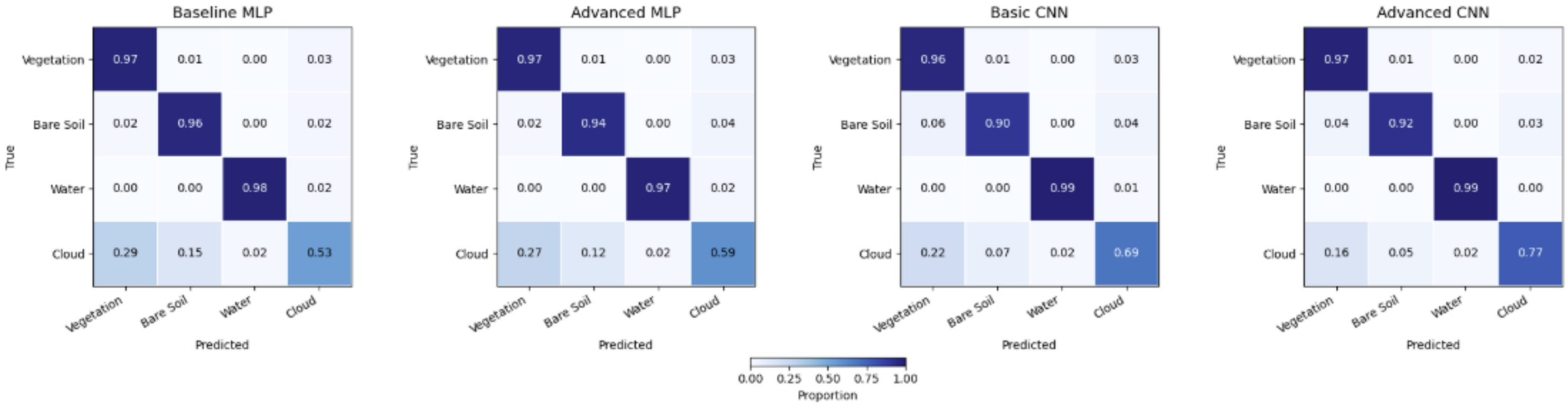




Test Results Comparison

- Validation findings hold consistently on unseen test data
- Performance hierarchy maintained across all models
- +3.7% improvement from Baseline MLP → Advanced CNN
- Cloud classification shows most significant gains
- CNN models demonstrate better generalization to new data

	Model	Approach	Test Accuracy	Cloud-P	Cloud-R	Parameters
0	Baseline MLP	Pixel-based	0.8952	0.813	0.534	2,916
1	Advanced MLP	Pixel-based	0.8989	0.794	0.587	45,124
2	Basic CNN	Patch-based	0.9047	0.819	0.687	56,084
3	Advanced CNN	Patch-based	0.9325	0.893	0.770	150,276



Key Insights & Future Work

Insights:

- Spectral-only models plateau at ~90% accuracy
- Adding spatial context (CNNs) boosts accuracy +3–4%
- Cloud detection benefits most (+24% recall improvement)
- Parameter efficiency varies across architectures

Future Work:

- Develop 10m resolution Scene Classification Layer to match Sentinel-2's finest spatial detail
- Extend pixel vs patch analysis across all Sentinel-2 resolutions (10m/20m/60m)
- Multi-temporal analysis for cloud detection
- Transfer learning for generalization

References

- Raiyani, K., Gonçalves, T., Rato, L., Salgueiro, P., & Marques da Silva, J. R. (2021). Sentinel-2 Image Scene Classification: A Comparison between Sen2Cor and a Machine Learning Approach. *Remote Sensing*, 13(2), 300. https://doi.org/10.3390/rs13020300
- Singh, R., Pal, M., & Biswas, M. (2025). Cloud detection methods for optical satellite imagery: A comprehensive review. *Geomatics*, 5(3), 27. https://doi.org/10.3390/geomatics5030027